PROJECT REQUIREMENTS

English Transcript to Indian Sign Language Gloss

Requirements specification, architecture, quality rules, and Phase 1 transcript-cleaning plan

Purpose
Define a clear, implementation-ready scope for converting English transcripts from prerecorded YouTube videos and livestreams into a clean, auditable text dataset that can later be translated into Indian Sign Language (ISL) gloss.

Project attribute	Current decision
Target sign language	Indian Sign Language (ISL)
Current source material	English transcripts only; sourced from prerecorded YouTube videos and livestreams
Possible future input	Gujarati transcripts may be added later; not part of the current Phase 1 deliverable unless explicitly requested
Current phase	Phase 1 — transcript inspection, cleaning, segmentation, and quality reporting
Later phase	English-to-ISL gloss translation
Explicitly out of scope for now	Gloss-to-sign generation, human signing-video retrieval, 3D avatar generation, and video overlay/synchronization
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Important design principle: preserve meaning and traceability. Cleaning must remove noise without rewriting the speaker's intended message or prematurely converting English into ISL.

1. Executive summary

The project will ultimately make video content more accessible by translating spoken-language transcripts into ISL and, in a later phase, displaying a signing representation alongside the original video. The current work is deliberately narrower: prepare trustworthy English text for translation.

The system must preserve the raw transcript, produce a cleaned version, record meaningful corrections, and retain timestamps and speaker information whenever available. It must not invent missing speech or silently remove words that could change meaning.

Phase 1 ends when the transcript corpus is clean, consistently formatted, traceable to its source, segmented into useful units, and accompanied by a quality report. No gloss translation is performed during Phase 1.

2. Goals and non-goals

Goals

- Accept English transcript files from prerecorded YouTube videos and livestreams.
- Normalize formatting and remove transcript artifacts while preserving spoken meaning.
- Retain timestamps, speaker labels, source identifiers, and uncertainty markers when available.
- Split text into sentence-like or meaning-preserving chunks suitable for later translation.
- Produce machine-readable outputs that another developer or AI tool can process without guessing the schema.
- Generate a cleaning report that records what changed, what was uncertain, and what requires human review.

Non-goals for the current phase

- Do not translate English into ISL gloss yet.
- Do not create, retrieve, or animate signs, signing videos, or 3D avatars.
- Do not synchronize signing with video playback.
- Do not use the transcript-cleaning step to paraphrase, summarize, simplify, or correct the speaker's argument.
- Do not assume a transcript's automatic captions are accurate merely because they are formatted correctly.

3. Scope and operating assumptions

ID	Requirement / assumption
S-01	ISL is the selected target sign language for later translation.
S-02	Current source language is English. Gujarati may be added in a later iteration.
S-03	Source material may come from prerecorded YouTube videos or livestreams.
S-04	The project currently has transcripts, not the underlying videos. Cleaning must work without video access.
S-05	Timestamps and speaker labels may or may not be present. Never fabricate them.
S-06	The raw input is immutable. All cleaning happens in a separate output.
S-07	Uncertain corrections must be flagged for review, not silently guessed.
S-08	The eventual translation must be meaning-based and ISL-aware; English words must not simply be replaced one by one with gloss labels.

4. End-to-end project architecture

Phase	Purpose	Output
Phase 1 — Transcript cleaning	Inspect, normalize, correct obvious artifacts, preserve meaning, segment, and report quality.	Clean transcript + metadata + change log + quality report
Phase 2 — English-to-ISL gloss	Translate the meaning of English text into a linguistically appropriate ISL gloss representation.	ISL gloss sequence + uncertainty flags + provenance
Phase 3 — Gloss validation	Have fluent ISL users/reviewers verify grammar, meaning, and completeness.	Reviewed reference/candidate glosses
Phase 4 — Sign generation (later)	Convert validated representation into a human signing video or animated avatar.	Signing media
Phase 5 — Video integration (later)	Place signing output in a corner of the source video and synchronize it.	Accessible video output

Phase 1 is the only implementation phase in scope now. Later phases are documented to guide data design, not to trigger their implementation.

5. Phase 1 functional requirements

ID	Requirement	Acceptance condition
P1-FR-01	Ingest plain-text transcripts (.txt) and, where practical, subtitle formats such as .srt and .vtt.	Input text is extracted; subtitle timestamps and cue structure are retained when available.
P1-FR-02	Keep a byte-for-byte or exact textual copy of the raw source before cleaning.	Raw source remains unchanged and can be traced to every cleaned output.
P1-FR-03	Normalize line endings, whitespace, encoding artifacts, and repeated formatting noise.	No accidental joining of words; normalized output is readable and consistent.
P1-FR-04	Handle subtitle timestamps and speaker labels as metadata, not as spoken words.	Metadata is preserved separately and not fed as transcript text.
P1-FR-05	Identify likely ASR artifacts such as duplicated phrases, malformed fragments, and obvious caption boilerplate.	Potentially meaningful repetitions are not deleted automatically without a rule or review.
P1-FR-06	Preserve meaning-bearing disfluencies, negation, uncertainty, emphasis, names, numbers, and technical terms.	No material semantic change caused by cleaning.
P1-FR-07	Segment long text into sentences or chunks while preserving order and source offsets/timestamps when available.	Every chunk can be mapped back to the original source.
P1-FR-08	Flag uncertain spans instead of inventing words.	Uncertain content has a review flag and remains visible.
P1-FR-09	Produce a machine-readable JSON output and a human-readable cleaned text output.	Outputs conform to the documented schema.
P1-FR-10	Produce a cleaning log and per-file quality report.	Corrections, warnings, counts, and review needs are recorded.

6. Transcript-cleaning policy

The cleaner should be conservative. Its job is to make text consistent and usable—not to make the speaker sound more polished than they were.

6.1 Safe automatic operations

- Normalize Unicode and whitespace without changing the spelling or meaning of words.
- Remove clearly non-spoken interface artifacts (for example, subtitle sequence numbers) from the text field while preserving them in metadata if useful.
- Standardize line breaks and merge subtitle cues into readable sentences where timing and sentence boundaries allow.
- Remove repeated headers/footers or explicit platform boilerplate only when confidently identified.
- Normalize obvious duplicated caption fragments only when the duplication is clearly an ingestion artifact and not actual repetition by the speaker.
- Preserve the original punctuation when possible; repair missing punctuation only as a formatting aid, without changing sentence meaning.

6.2 Operations requiring caution or review

- Automatic speech-recognition substitutions, especially names, acronyms, domain terms, numbers, and words that sound alike.
- Repeated words and phrases: they may be stutters, emphasis, turn-taking, or genuine content.
- Fillers such as “um”, “uh”, “you know”, and “like”: do not remove by default. Their removal must be configurable and recorded because they can affect discourse and timing.
- Incomplete sentences, interruptions, overlapping speakers, and false starts.
- Profanity, dialect, non-standard grammar, code-switching, and regional vocabulary.
- Unclear audio markers such as [inaudible]. Preserve them and flag them; do not guess the missing speech.

6.3 Prohibited operations

- Do not summarize, paraphrase, simplify, translate, or reorder statements.
- Do not delete negation, hedging, qualifiers, names, dates, numbers, or technical vocabulary just to make the text cleaner.
- Do not fabricate missing timestamps, speaker identities, words, or source metadata.
- Do not silently resolve ambiguous text using outside knowledge.
- Do not convert English into ISL gloss during cleaning.

7. Recommended processing workflow

1. **Register the file.** Assign a stable file ID and record original filename, source type, input format, language, and processing date if known.
2. **Preserve the raw input.** Save the original file unchanged and compute a checksum if the implementation supports it.
3. **Parse the format.** Extract text from TXT/SRT/VTT. Keep timestamps, cue numbers, and speaker labels in separate fields.
4. **Normalize.** Fix encoding, whitespace, line endings, and formatting artifacts.

5. **Detect candidate issues.** Identify duplicated text, likely ASR errors, boilerplate, malformed fragments, and uncertain spans.
6. **Apply conservative rules.** Automatically fix only high-confidence formatting or ingestion issues. Log all substantive edits.
7. **Segment.** Divide the cleaned transcript into sentence-like units or short context-preserving chunks. Preserve ordering and source links.
8. **Validate.** Run checks for empty outputs, accidental word concatenation, dropped content, duplicate chunks, invalid timestamps, and suspiciously large text loss.
9. **Export.** Write clean text, structured JSON, edit log, and a quality report.
10. **Review exceptions.** Put uncertain or potentially meaning-changing cases into a review queue.

8. Required output format

For each source transcript, generate at least four artifacts. Use consistent file naming based on the source ID.

Artifact	Purpose	Suggested filename
Raw source	Original input, unchanged.	source_id.raw.ext
Clean transcript	Human-readable cleaned English text.	source_id.cleaned.txt
Structured transcript	Chunks, timestamps, speaker labels, offsets, flags, and provenance.	source_id.cleaned.json
Cleaning report	Summary of actions, counts, warnings, and review items.	source_id.cleaning_report.json

8.1 Suggested JSON schema

```
{ "schema_version": "1.0", "source_id": "stable-id", "source_filename": "original-file-name",
"source_type": "youtube_prerecorded | livestream | unknown", "source_language": "en",
"timestamps_available": true, "cleaning_status": "complete | needs_review | failed", "chunks": [ {
"chunk_id": "stable-id-0001", "text": "Cleaned sentence or context-preserving chunk.", "start_time":
"00:00:12.400", "end_time": "00:00:15.200", "speaker_id": null, "source_start_offset": 0,
"source_end_offset": 58, "uncertainty_flags": [], "edit_notes": [] } ], "warnings": [],
"review_required": false }
```

Schema rules

- Use null for unavailable timestamps or speaker IDs; never make up values.
- Use timestamps in a consistent format, preferably HH:MM:SS.mmm when present.
- Offsets should refer to a documented source representation; if offsets cannot be maintained reliably, omit them rather than output incorrect values.
- Chunks must preserve transcript order. Text in metadata fields must not be mixed into the spoken text.
- The schema may be extended, but field names and meanings must remain documented and stable.

9. Quality checks and acceptance criteria

Check	Acceptance criterion
Traceability	Every cleaned file has a stable source ID and points back to its raw input.
Meaning preservation	No unlogged content removal or semantic rewriting.
Timestamp integrity	All retained timestamps parse correctly; missing timestamps remain null.
Speaker integrity	Speaker labels are preserved when available and never invented.
Segmentation integrity	Chunks remain in order and no words are lost or duplicated by the segmenter.
Auditability	Substantive changes and review flags are recorded.
Machine readability	JSON parses successfully and follows the schema.
Human readability	Cleaned text is readable, with words separated and sensible line breaks.
Conservative handling	Uncertain text is flagged rather than guessed.
No scope creep	No gloss generation or signing generation occurs during Phase 1.

10. Cleaning examples

Input	Desired Phase 1 output	Reason
SRT cue: 17 / 00:00:10,000 --> 00:00:12,000 / We we need to test it.	Text: “We we need to test it.” Keep cue timing as metadata; flag repetition if appropriate.	Do not assume repetition is a caption bug; it may be spoken.
[inaudible] and then we deployed the model.	Preserve “[inaudible] and then we deployed the model.” Add an uncertainty flag.	Never invent missing speech.
uh, the model is not ready yet	Keep the sentence and filler by default; normalize spacing and punctuation if needed.	“Not” is critical; fillers are configurable, not silently removed.
Speaker 2: The result was 15, not 50.	Preserve the words and number; keep speaker label separately.	Numbers and negation must not be altered.
Caption line 1: “We are building” / line 2: “a translation system.”	Merge into one sentence if no meaningful boundary is lost; preserve timing metadata for the relevant span.	Subtitle cues are not always sentence boundaries.

11. Implementation instructions for the coding tool

The following instructions are intended to be directly actionable by a developer or coding agent.

- Implement Phase 1 only. Do not implement gloss translation or signing generation unless separately requested.
- Build a deterministic cleaning pipeline first. Keep any LLM-assisted correction optional, disabled by default, and reviewable.
- Separate parsing, normalization, issue detection, conservative cleaning, segmentation, validation, and export into distinct functions/modules.
- Make cleaning rules configurable. For example, filler removal must be an explicit option and must be off by default.
- Keep the raw input immutable. Never overwrite the user's source transcript.
- Log changes at the smallest practical span: original text, cleaned text, operation, confidence/reason, and whether human review is required.
- When confidence is low, preserve the original wording and emit a warning.
- Add unit tests for timestamps, repeated phrases, speaker labels, numbers, negation, unclear audio, Unicode, empty files, and long transcripts.
- Provide a command-line or simple batch interface that can process one file or a folder and report per-file success/failure.
- Ensure one bad file does not silently stop the batch; report failures and continue when safe.
- Do not claim a transcript is accurate against the original audio because the source videos/audio are not currently available.

12. Suggested folder structure

```
project/ data/ raw/ # immutable originals cleaned/ # human-readable cleaned text structured/ # JSON with chunks and metadata reports/ # per-file and batch reports review_queue/ # uncertain spans requiring review src/ ingest/ # TXT/SRT/VTT parsing normalize/ # whitespace and encoding normalization detect/ # artifact and uncertainty detection clean/ # conservative cleaning rules segment/ # sentence/chunk segmentation validate/ # integrity and schema checks export/ # TXT/JSON/report writing tests/ README.md
```

13. Requirements for the later English-to-ISL gloss phase

These requirements are recorded for continuity only; they are not part of Phase 1 implementation.

- The translator must translate sentence meaning into ISL, not substitute one English word with one gloss label.
- Gloss is a written representation/annotation of signs; it is not the sign itself and does not fully encode all facial, body, spatial, and timing information.
- Candidate resources previously identified include Virtual ISL Interpreter (rule-based), Yap & Render (LLM plus rules), and SignAI (structured LLM output). Each must be tested rather than assumed correct.
- The IIIT Bangalore project describes a small English-to-ISL gloss parallel corpus; availability and terms must be verified before use.
- ISLTranslate is primarily ISL-video-to-English and should not be treated as a ready-made English-to-ISL gloss dataset without verifying the annotations.
- Every model or rule-based output needs review by fluent ISL users. Automated plausibility is not a substitute for linguistic validation.
- Store source sentence, candidate gloss, method/model version, uncertainty, and reviewer feedback together.

14. Risks and mitigations

Risk	Mitigation
ASR errors remain in source	Flag suspicious words and state clearly that audio verification is unavailable.
Over-cleaning changes meaning	Use conservative rules, preserve raw data, and maintain edit logs.
Timestamps are lost during segmentation	Represent timing separately and test round-trip parsing.
Different sources use different formats	Normalize to a common internal schema after format-specific parsing.
LLM invents corrections	Disable automatic generative rewriting by default; require review for uncertain text.
Gloss output is not natural ISL	Use a fluent ISL reviewer and a reference set before scale-up.
Future Gujarati input complicates design	Keep language metadata explicit and isolate language-specific processing; do not mix Gujarati into the English Phase 1 batch unless instructed.

15. References and project resources

These are starting points identified during project research. Verify current repository status, code dependencies, data access, and licensing before adopting them.

- Virtual ISL Interpreter (IIIT Bangalore-related prototype): github.com/krishnshyam/VirtualISLInterpreter
- Yap & Render: github.com/mitarthpathak/yap-render
- SignAI — Speech and Text to Indian Sign Language: GitHub repository
- ISLTranslate: github.com/Exploration-Lab/ISLTranslate
- RKMVERI ISL project and dictionary: cs.rkmvu.ac.in/~isl

16. Phase 1 completion checklist

- Raw input files are preserved and identifiable.

- Transcript formats are parsed without losing available timing or speaker metadata.
- Cleaned English text is exported without semantic rewriting.
- Text is segmented into ordered, traceable chunks.
- Structured JSON passes schema validation.
- Cleaning logs and quality reports are generated.
- Uncertain or suspicious spans are flagged for human review.
- No English-to-ISL gloss translation or sign-generation work has been performed.
- The user has reviewed a sample output and approved the cleaning policy before large-scale processing.

End of specification. Implement Phase 1 only, then report outputs, warnings, and unresolved decisions before proceeding to English-to-ISL gloss translation.